

Math 290

Practice Quiz 2 — Version C Solutions

1(a) Answer: E. The proof asserts $k/3 \in \mathbb{Z}$ with no reason. (It happens to be true, but proving it takes as much work as the theorem itself — this is exactly the warning after Proposition 6.19 about dividing.) So $n = 2(k/3)$ does not yet show n is even. A correct direct proof avoids dividing: $n + 7 = 3n - 2n + 7 = 2k - 2n + 7 = 2(k - n + 3) + 1$, which is odd since $k - n + 3 \in \mathbb{Z}$. (Choice D is not a flaw: in a direct proof you are *supposed* to assume the premise “ $3n$ is even.”) (*Based on Proposition 6.19.*)

1(b) Answer: True. $x^2 - 2x \geq -1 \iff x^2 - 2x + 1 \geq 0 \iff (x - 1)^2 \geq 0$, which holds for every real x . The conclusion is always true, so the implication is trivially true. (*Based on Proposition 6.9.*)

1(c) Answer: False. The inverse $\neg P \Rightarrow \neg Q$ is the contrapositive of the converse, so it has the same truth table as the *converse* $Q \Rightarrow P$, not as $P \Rightarrow Q$. (Row P false, Q true: $P \Rightarrow Q$ is true, the inverse is false.) (*Based on Section 7.C.*)

2. (*Based on Exercise 8.1.*)

Solution. Let $x, y \in \mathbb{Z}$. We work directly. Assume x and y have the same parity. There are two cases. Case 1: x and y are both even, say $x = 2k$, $y = 2\ell$ with $k, \ell \in \mathbb{Z}$. Then $x^2 + xy = 4k^2 + 4k\ell = 2(2k^2 + 2k\ell)$, which is even. Case 2: x and y are both odd, say $x = 2k + 1$, $y = 2\ell + 1$ with $k, \ell \in \mathbb{Z}$. Then $x + y = 2(k + \ell + 1)$, so $x^2 + xy = x(x + y) = 2((2k + 1)(k + \ell + 1))$, which is even. In every case $x^2 + xy$ is even. $\square$

3. (*Based on Exercise 9.3 (with Exercise 8.3).*)

Solution. Let $a, b, c \in \mathbb{Z}$ with $a^2 + b^2 = c^2$. Assume, by way of contradiction, that a and b are both odd. Write $a = 2k + 1$ and $b = 2\ell + 1$ with $k, \ell \in \mathbb{Z}$. Then

$$c^2 = a^2 + b^2 = (4k^2 + 4k + 1) + (4\ell^2 + 4\ell + 1) = 4(k^2 + k + \ell^2 + \ell) + 2,$$

so $c^2 \equiv 2 \pmod{4}$. But $c^2 \equiv 0$ or $c^2 \equiv 1 \pmod{4}$. If $c^2 \equiv 0$, then $4 \mid c^2$ and $4 \mid (c^2 - 2)$, so 4 divides their difference 2, which is false. If $c^2 \equiv 1$, then $4 \mid (c^2 - 1)$ and $4 \mid (c^2 - 2)$, so $4 \mid 1$, which is false. Either way we reach a contradiction. Hence a is even or b is even. $\square$

4. (*Based on Exercise 10.8 (the inclusion there is actually an equality).*)

Solution. Let x be arbitrary. Then

$$\begin{aligned} x \in A - (B \cap C) &\iff x \in A \wedge x \notin B \cap C && \text{(definition of set difference)} \\ &\iff x \in A \wedge \neg(x \in B \wedge x \in C) && \text{(definition of intersection)} \\ &\iff x \in A \wedge (x \notin B \vee x \notin C) && \text{(De Morgan's law)} \\ &\iff (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) && \text{(distributive law for } \wedge \text{ over } \vee) \\ &\iff x \in A - B \vee x \in A - C && \text{(definition of set difference)} \\ &\iff x \in (A - B) \cup (A - C) && \text{(definition of union).} \end{aligned}$$

Since $x \in A - (B \cap C) \iff x \in (A - B) \cup (A - C)$ for every x , the sets are equal (definition of set equality). $\square$